Do frozen geospatial foundation-model embeddings detect surface change better than spectral indices? A noise-normalised comparison on forest loss and Himalayan glacial lakes

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Abstract

Foundation models pretrained on satellite imagery are promoted for change detection but rarely compared with simple spectral indices under a common standard. We score reflectance, spectral indices, a signed NDVI-decrease baseline and two foundation-model embeddings (Prithvi-EO-2.0 and an RGB-only SSL4EO-S12 ResNet-50) with one rule: the change between two periods in units of the scene-to-scene noise, which works for feature vectors of any length. We evaluate against separate reference products: forest loss in five Indian areas (Hansen Global Forest Change, which shares optical signal with the index baselines, in areas chosen for active clearing) and water change at four Nepal glacial lakes (Sentinel-1 radar). On forest loss no foundation model beats the unsupervised signed NDVI baseline in any area; Prithvi is not distinguishable from it in two (average precision 0.69 against 0.72 in Bastar, 0.52 against 0.54 in Kodagu) and is clearly worse in three. A trained linear readout of Prithvi beats the baseline in two areas, but a readout of six raw bands does about as well. On glacial lakes Prithvi (0.63) and raw bands (0.64) beat the MNDWI water index (0.45) and tie each other, but lake by lake the advantage is individually clear only at one lake, where MNDWI is at chance; at Imja Tsho, which holds 34 of 59 positive patches, none differ clearly, and no lake holds enough spatial blocks to settle the question. In a placebo test on twelve areas the embeddings raise 0 to 3.4% false alarms against up to 41% for indices, a property of within-period similarity rather than detection skill. Code, scene lists and checksums are released.

1 Introduction

Detecting where a landscape has changed between two dates is a basic task in remote sensing, and foundation models trained on millions of Sentinel-2 scenes [1], [2] are now offered as a general-purpose answer. Evidence that they beat the long-standing alternative, a spectral index such as NDVI thresholded on its difference [3], [4], [5], is thin, usually reported on benchmarks [6], [7] rather than against independent reference data, and almost never with the noise between acquisitions taken into account. Two images of the same place never agree exactly: clouds, haze, sun angle and season change every pixel. A method that reports "change" without a notion of this noise cannot say how much of its output is real.

We ask a narrow question: when every method is judged against the same scene-to-scene noise and the same reference data, do foundation-model embeddings detect real surface change more reliably than spectral indices, and does the answer hold across fields? We contribute (i) a noise-normalised change score for any feature vector, (ii) an evaluation in two fields (five forest areas, four glacial lakes) whose scoring design was fixed before any score was computed for the added forest areas (the earlier areas and the lakes are covered in Section 3.2 and the Reproducibility paragraph), with spatial block bootstrap intervals and a reference-free placebo test, and (iii) a negative-leaning result reported as such: no foundation model beat the best simple baseline when scored without labels, and a trained linear readout of the frozen features, tried afterwards, did no better than the same readout of raw reflectance.

2 Data and reference products

Optical. Sentinel-2 L2A scenes from the Element 84 Earth Search catalogue, six bands (blue, green, red, narrow NIR, SWIR1, SWIR2), cloud and snow masked with the scene classification layer. Forest periods use three clear scenes each inside one February to March window. At three lakes the earlier period had one clear scene and the later three, so their noise scale comes from the later period alone; the fourth lake has three in both.

Forest loss. Five areas of about 7.6 by 8 km, compared between February to March 2021 and 2025: Bastar and Odisha (central-east India), chosen first, and Kodagu and Wayanad (Western Ghats), Garo Hills (Meghalaya) and the Shivalik foothills (Uttarakhand), added afterwards to test whether the Bastar and Odisha result held in other forest types. The three added areas were registered before any of their scores was computed (`studies/forest_expansion_prereg.yaml`), but after the Bastar and Odisha results were known. Boxes centred on a stated point failed a label-only screen of at least 150 positive patches, so we amended the registration, still before any scene or score: each box is the one with most positive patches inside a 0.4 degree window, chosen from Hansen labels alone. A fourth region, the Nepal Terai near Chitwan, failed even with its best box (40 positives) and is not evaluated. The boxes therefore overrepresent active clearing, which probably favours a greenness-drop score. The reference is Hansen Global Forest Change v1.12 [8], which is built from Landsat optical reflectance and so shares signal with NDVI, an advantage for the index baselines: a patch is positive if at least 10% of it was lost in 2022 to 2024 and clean if under 2% was lost in 2021 to 2024; patches between are set aside, and only patches with at least 50% forest cover are eligible.

Glacial lakes. The reference is the change in Sentinel-1 water frequency per patch (Sentinel-1 RTC from Microsoft Planetary Computer, scenes within 20 days of each window, usable orbits only). A patch is positive if at least 25% of it changed water state and clean if under 5% did. Radar is a different sensor from the optical data under test, so the reference is not circular, but it has its own errors. Three pilot lakes (Tsho Rolpa, Imja Tsho, Thulagi; October to November 2018 against 2024) held 44 positive patches, 34 at one lake. We therefore screened all 30 lakes of our Hi-MAG-based inventory [9] with the radar reference alone, before any optical scene or score, and added each other lake with at least 10 positive patches (`studies/lake_expansion_amendment.yaml`). Two passed, but neither had a clear 2018 optical scene, so the earlier period was moved to 2020 for added lakes; the repeated screen kept one (15 positives) and dropped the other (7). The added lake was chosen because the radar shows change at it; Tsho Rolpa and Thulagi, where radar shows little change, were pilot lakes fixed before the screen. The pooled set mixes a 2018 earlier period with one clear scene at three lakes and a 2020 earlier period with three scenes at the fourth, so z ranks patches whose noise scale was estimated differently.

All references are other satellite products and are treated as agreement targets, not ground truth.

3 Methods

3.1 A change score in units of noise

Each scene is reduced to one feature vector per 16×16 pixel patch (160 m). Let $\bar{f}_A$ and $\bar{f}_B$ be the mean feature vectors over the n_A and n_B scenes of the earlier and later period, and $D^2 = \|\bar{f}_B - \bar{f}_A\|^2$. The noise is estimated from pairs of scenes in the same period,

$$\sigma^2 = \mathrm{mean}_{i<j} \ \tfrac{1}{2}\|f_i - f_j\|^2,$$

pooled over a 3×3 patch neighbourhood because three scenes give only three pairs. The score is

$$z^2 = \frac{D^2}{\sigma^2\,(1/n_A + 1/n_B)},$$

about 1 when nothing has changed between the compared scenes, whatever the length or scale of the features; Section 4.3 checks whether it stays near 1 between years. The HabitatShift pipeline [10] already tests whether the change at a pixel exceeds the scene-to-scene spread of its own period; this score is that test written for feature vectors of any length.

3.2 Methods and evaluation

All methods use the same patches and the same z: single indices (NDVI, NDWI, MNDWI, NDBI), five indices together, six standardised raw bands, *Prithvi-EO-2.0-300M* [1] (encoder only, single time step, six bands, 1,024 numbers per patch) and an *SSL4EO-S12 ResNet-50* [2] using layer-3 features of the RGB bands only. Because z is unsigned, forest loss also gets a *signed NDVI-decrease* baseline, which uses the sign of the NDVI change and which an unsupervised embedding distance cannot provide. Prithvi takes six-band surface reflectance scaled by the model's own band means and standard deviations, one scene at a time (a single time step, so every scene has its own embedding and the scene-to-scene spread can be measured); it was pretrained on Harmonized Landsat and Sentinel-2 data at 30 m [1], whereas our Sentinel-2 pixels are 10 m. The ResNet takes red, green and blue reflectance divided by 10,000. No correction is made for the many paired intervals reported; each is a separate 95% interval. The scoring design was fixed before any score was computed for the added forest areas; for Bastar, Odisha and the pilot lakes, which were chosen earlier without a written selection rule, the scripts and their results were committed together (`d5ebab4`), so there the order rests on our statement. We report average precision (AP) and ROC-AUC against the reference labels, with 95% intervals from a spatial block bootstrap (blocks of 8×8 patches, 500 resamples) and paired differences from a named baseline resampled on the same blocks. In the placebo test one scene of a period is compared with the rest of the same period, with the noise estimated without the held-out scene; nothing has changed, so every patch with $z > 3$ is a false alarm. This needs no reference and was run on twelve areas.

4 Results

4.1 Forest loss: no foundation model beats a signed NDVI baseline

Figure 1 and Table 1 show AP against Hansen loss. No foundation model beats the signed NDVI-decrease score in any area. Prithvi is not distinguishable from it in Bastar (0.69 against 0.72) and Kodagu (0.52 against 0.54) and clearly worse in Odisha (0.56 against 0.76), Garo Hills (0.48 against 0.55) and the Shivalik foothills (0.64 against 0.77). The RGB-only SSL4EO model is clearly worse in four areas. Six raw bands tie the baseline in three areas and are far below it in Odisha and the Shivalik foothills. Prithvi is not distinguishable from raw bands in four areas (paired intervals include zero; Odisha 0.56 against 0.56) and beats them only in the Shivalik foothills (+0.35 to +0.52), where unsigned raw-band scores are near chance (AP 0.19, AUC 0.47); the leaf state of deciduous sal forest in February to March can differ between years, which we did not test. The comparison is partly about direction: against unsigned NDVI (0.62, 0.53, 0.53, 0.28, 0.12) Prithvi has the higher AP in four areas and is level in Kodagu (no paired interval was computed). Giving the unsigned scores the same direction (exploratory), by zeroing z wherever NDVI did not fall, lifts Prithvi to 0.77, 0.72, 0.54, 0.51, 0.64 and raw bands to 0.76, 0.73, 0.55, 0.53, 0.63: Prithvi is then ahead of signed NDVI in Bastar (+0.03 to +0.10), level in Kodagu and behind in Odisha, Garo Hills and the Shivalik foothills, and within 0.02 AP of the gated raw bands in every area. Direction closes part of the gap, but not through anything the embeddings add. At the fixed rule $z > 3$ the embeddings catch more true loss but flag many more clean patches (Bastar: Prithvi

48% recall at 5.6% false positives, signed NDVI 24% at 0.6%); at a matched 5% false-positive rate the recall of signed NDVI (0.58, 0.60, 0.62, 0.33, 0.67) is higher than Prithvi's (0.46, 0.23, 0.55, 0.21, 0.47) in every area (point estimates; no intervals were computed). Signed NDVI also matches or beats both embeddings in 13 of 15 variants that define positives from 2021 to 2024 or move the cut-off to 5% or 20% (the exceptions are Bastar and Garo Hills with positives from 2021, where Prithvi is ahead by 0.03 and 0.02 AP); no paired intervals were computed for these variants.

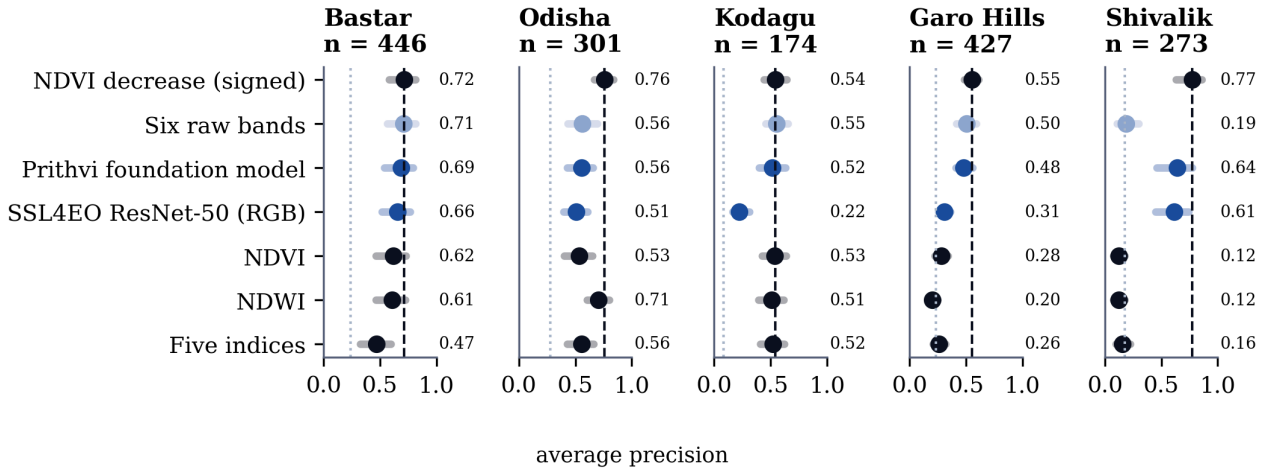


Figure 1. Average precision against Hansen forest loss, with 95% block-bootstrap intervals. Blue: foundation models; light blue: raw bands; dark: spectral indices. Dashed line: signed NDVI-decrease baseline; dotted line: chance.

Table 1. Forest loss in five areas, average precision (ROC-AUC). The signed NDVI-decrease score is the baseline; * means the paired difference from it, resampled on the same spatial blocks, has a 95% interval that excludes zero (every such difference is negative, so the method is worse). Positive patches: 446, 301, 174, 427 and 273.

Score	Bastar	Odisha	Kodagu	Garo Hills	Shivalik
NDVI decrease (signed)	0.72 (0.84)	0.76 (0.84)	0.54 (0.92)	0.55 (0.78)	0.77 (0.90)
Six raw bands	0.71 (0.85)	0.56 (0.76) *	0.55 (0.93)	0.50 (0.73)	0.19 (0.47) *
Prithvi-EO-2.0	0.69 (0.88)	0.56 (0.80) *	0.52 (0.91)	0.48 (0.78) *	0.64 (0.90) *
SSL4EO ResNet-50 (RGB)	0.66 (0.83)	0.51 (0.75) *	0.22 (0.76) *	0.31 (0.66) *	0.61 (0.89) *
NDVI (unsigned)	0.62 (0.78) *	0.53 (0.73) *	0.53 (0.92) *	0.28 (0.54) *	0.12 (0.32) *
NDWI	0.61 (0.74) *	0.71 (0.83) *	0.51 (0.90) *	0.20 (0.41) *	0.12 (0.31) *
Five indices	0.47 (0.64) *	0.56 (0.73) *	0.52 (0.91)	0.26 (0.52) *	0.16 (0.44) *

4.2 Glacial lakes: the pooled result favours models and raw bands over the water index, unevenly

Figure 2 and Table 2 give the lake result with paired intervals against MNDWI. Prithvi (AP 0.63) and six raw bands (0.64) beat MNDWI (0.45), with intervals that exclude zero (+0.06 to +0.22 and +0.06 to +0.24); the RGB-only model, NDVI and the five indices do not. Prithvi and raw bands are indistinguishable, so the gain over MNDWI is not evidence that pretraining helps. In point estimates z scores higher than the raw distance for raw bands (0.64 against 0.20) and Prithvi (0.63 against 0.35) but not the RGB-only model (0.43 against 0.70; z was the pre-specified score); no paired interval was computed. The result rests on 59 positive patches in only 11 spatial blocks: 34 at Imja Tsho, which grew from about 1.1 to 1.7 km^2, 15 at the added lake and 10 at the two lakes that stayed about the same size, so the intervals are wide (Prithvi AP 0.63, 95% interval 0.22 to 0.85). Taken lake by lake (Table 3, exploratory, run after the pooled result) the advantage over MNDWI is not general. At Imja Tsho alone Prithvi (0.80) and raw bands (0.81) are not distinguishable from MNDWI (0.60; paired intervals -0.16 to +0.27 and

-0.18 to +0.27). At the added lake MNDWI is at chance (AP 0.05, AUC 0.44) while the embedding and band scores are well above it, with wide intervals. Without Imja Tsho the direction holds with wide intervals (raw bands +0.01 to +0.37, Prithvi +0.004 to +0.25, RGB model +0.06 to +0.35). The advantage over MNDWI appears in the point estimates at three of the four lakes and in the three pilot lakes alone (Prithvi +0.02 to +0.23), but for Prithvi and raw bands it is individually clear only at the added lake, where MNDWI fails for reasons we did not examine (the RGB model's interval also excludes zero at Thulagi, on two blocks); at Tsho Rolpa, with four positives, MNDWI, raw bands and Prithvi are near or below chance, which may be reference noise. No lake holds enough spatial blocks to settle the question. MNDWI is also a weak baseline at turbid, partly iced lakes (the optical rule's Imja Tsho area was 27% below the published outline), so part of the gap is the baseline failing rather than the models succeeding. The reference itself is also disputed at Imja Tsho: a radar water model trained on optical labels, described in the technical report [10], keeps its area near 1.0 to 1.3 km^2 while the radar threshold used as the reference rises to 1.7 km^2.

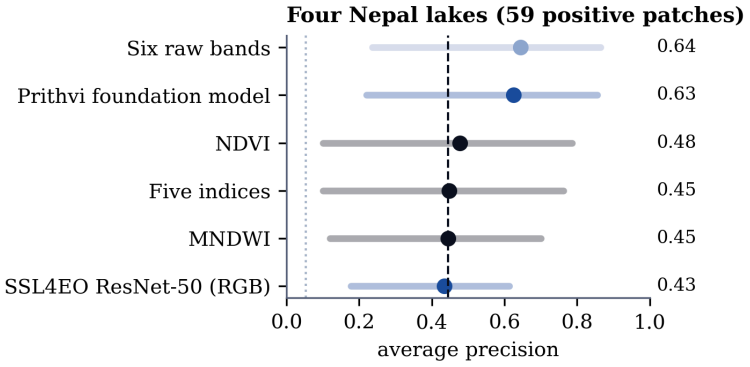


Figure 2. Average precision against the radar water-change reference at four Nepal lakes.

Table 2. Glacial lakes, average precision (ROC-AUC) and paired difference from MNDWI (95% interval; * excludes zero). 59 positive and 1,037 clean patches.

Score	AP (AUC)	Paired diff. from MNDWI
Six raw bands	0.64 (0.90)	+0.06 to +0.24 *
Prithvi-EO-2.0	0.63 (0.88)	+0.06 to +0.22 *
NDVI	0.48 (0.73)	-0.07 to +0.11
MNDWI	0.45 (0.76)	baseline
Five indices	0.45 (0.75)	-0.07 to +0.10
SSL4EO ResNet-50 (RGB)	0.43 (0.87)	-0.23 to +0.23

Table 3. Lakes one at a time, average precision of the noise-normalised score (positive patches, with the number of 8 x 8 patch spatial blocks that contain them). Exploratory.

Lake or group	Positives (blocks)	MNDWI	Six raw bands	Prithvi-EO-2.0	SSL4EO RGB
Imja Tsho	34 (4)	0.60	0.81	0.80	0.55
Added lake	15 (3)	0.05	0.64	0.56	0.70
Thulagi	6 (2)	0.84	0.90	0.93	1.00
Tsho Rolpa	4 (2)	0.01	0.04	0.04	0.13
Three pilot lakes (2018 against 2024)	44 (8)	0.57	0.76	0.75	0.45
All lakes except Imja Tsho	25 (7)	0.23	0.41	0.34	0.41

4.3 Placebo test and between-year noise: embeddings are quiet within a period, not between years

In the placebo test on twelve areas (the five forest areas, Chilika in two seasons, Coringa, Langtang, Purepu, Kathmandu and the Western Ghats; Figure 3) the two embedding models raise 0 to 3.4% false alarms at $z > 3$ (medians under 0.5%), whereas indices and raw bands raise up to 41% (medians 3 to 13%). This is not better detection: on Bastar, NDVI flags 23% of patches in the placebo test but only 1% of Hansen's clean patches, so the placebo rate depends on how alike the noise is in the two periods. A likely reading is that embeddings vary little between scenes taken days apart but respond to year-to-year differences (season, illumination, phenology) that the references do not count as change. The between-year check points the same way: on Hansen clean patches, where no forest was lost (other change may have occurred), the median z across the five forest areas is 0.7 to 1.5 for raw bands and 0.5 to 2.0 for the five indices, but 1.5 to 2.3 for Prithvi and 1.7 to 2.8 for the RGB model, and 3% to 20% of Prithvi's clean patches (RGB model: 3% to 37%) exceed $z > 3$. The five indices in Odisha show the same inflation (median 2.0, 15% above $z > 3$), so it is not unique to embeddings. Within-period noise therefore understates the between-year spread for the embeddings and for some indices, so z is not a calibrated test between years and is best read as a per-patch normalisation.

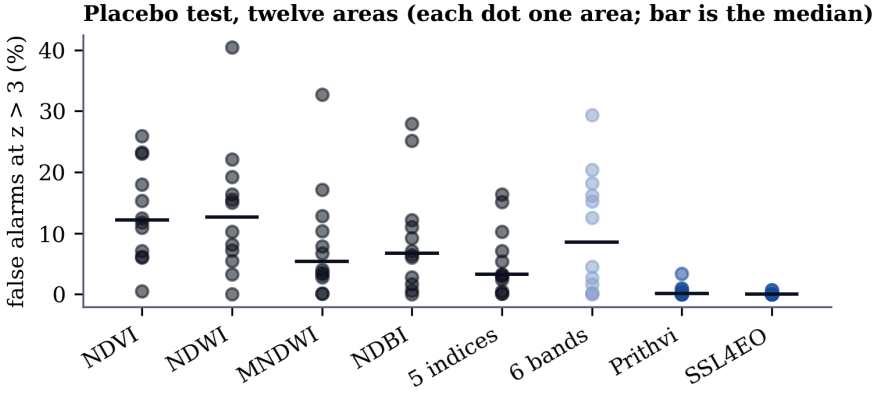


Figure 3. False alarms at $z > 3$ when one scene is compared with the rest of its own period, twelve areas.

4.4 A trained readout of the frozen features (exploratory)

A reviewer can fairly ask whether embeddings do better when read out with labels. After the main evaluation we trained an L2-regularised logistic regression on the signed noise-normalised change vector (the change in the feature vector over its noise scale, so its length is z), leaving one forest area out in

turn and choosing the regularisation only inside the four training areas (`studies/probe_design.yaml`, written before the run). A probe on six raw bands is the control. The paired comparison with the six-band probe and a wider regularisation grid were added after the first run showed they were needed, and are exploratory. Table 4 gives the result.

The Prithvi probe beats the unsupervised signed NDVI baseline in Bastar and the Shivalik foothills and ties it elsewhere, so a trained readout recovers an advantage that unsupervised z lacks. But the six-band probe does about as well: with the original grid it is better than the Prithvi probe in Kodagu and, narrowly, Bastar, and the Prithvi probe is ahead only in the Shivalik foothills, and neither is clearly ahead in any area under both grids. The RGB-only probe is worse than the baseline in four areas, and the probe on the five indices, a second control, is worse in Bastar, Odisha, Garo Hills and the Shivalik foothills on the original grid and not better anywhere. For Prithvi, the RGB model and the six-band probe the chosen regularisation sat at the lowest grid value in every fold (the five-indices probe chose 0.01 in two); with a wider grid the Prithvi probe beats the baseline in the same two areas and, only just (lower bound +0.0001), in Odisha, and the comparison with the six-band probe stays unclear. Pretraining is therefore not shown to help, though five folds whose pairwise results shift with the grid cannot show that it does not. A pooled leave-one-lake-out probe on the four lakes was not robust (Prithvi probe AP 0.77 against 0.61 for six bands on the original grid, 0.54 against 0.39 on the wider grid, intervals spanning zero in one case) and is not read as evidence.

Table 4. Linear probe on forest loss, average precision of the held-out area (difference with 95% block-bootstrap interval; * excludes zero). The last column repeats the paired comparison with the wider regularisation grid.

Area	Signed NDVI (no labels)	Prithvi probe	Prithvi probe minus signed NDVI	Six-band probe	Prithvi probe minus six-band probe	Same, wider grid
Bastar	0.72	0.77	+0.02 to +0.11 *	0.81	-0.07 to -0.00 *	-0.04 to +0.01
Odisha	0.76	0.79	-0.01 to +0.07	0.76	-0.00 to +0.08	-0.01 to +0.05
Kodagu	0.54	0.49	-0.11 to +0.03	0.55	-0.12 to -0.01 *	-0.09 to +0.01
Garo Hills	0.55	0.57	-0.03 to +0.06	0.59	-0.06 to 0.00	-0.06 to -0.00 *
Shivalik	0.77	0.85	+0.04 to +0.13 *	0.80	0.00 to +0.13 *	-0.01 to +0.09

5 Discussion and limitations

What the evidence supports. For forest loss, a signed NDVI-decrease score matches or beats Prithvi and a smaller RGB model in every area. For lake water change, Prithvi beat the standard water index in the pooled result but not plain reflectance, and the advantage is individually clear at only one of four lakes. We found no setting in which an embedding clearly beat the best simple baseline: unsupervised scores never did, and a trained readout of Prithvi, which beat the unsupervised baseline in two forest areas, was matched by a readout of raw bands.

What it does not. Five forest areas, three chosen by a label-only search for active clearing, and two substantially changing lakes, one of them chosen because radar shows change at it, are too few to settle the question; three scenes per period is a small noise sample; and urban or wetland change, which needs other reference products, was not tested. An unsupervised embedding score gives no direction of

change, a real limit for loss detection that the probe sidesteps by seeing the sign. Every reference is another model's output, so all scores are agreement, not accuracy. Fine-tuning, the usual way to deploy such models, was not tested: our claim is limited to frozen single-date embeddings of these two models, and the SSL4EO model sees RGB only. There is one forest reference (Hansen) and no hand-checked sample. A radar water model with conformal lake-area intervals, built on the same lakes, is described in the technical report [10].

6 Conclusion

Under a common noise normalisation, foundation-model embeddings did not outperform a signed vegetation baseline on forest loss, and their advantage over a water index at glacial lakes was matched by raw reflectance and individually clear at only one of four lakes. A linear probe on Prithvi recovers an advantage over the unsupervised baseline in two areas, but a probe on six raw bands does about as well, so pretraining is not shown to help. The embeddings' low false-alarm rate in a placebo test is a property of within-period similarity, not evidence of detection skill. We release the pipeline, the evaluation design and all recorded outputs so that others can test further fields and fine-tuned models under the same standard.

Reproducibility. Code, scene lists, weight checksums and the exact numbers behind every table are in the project repository (`docs/V3_RESULTS.md`, `data/experiments/`), archived with a DOI [10]. The forest registration, the label screen, the amendment and the box selection are commits `9fc28c3`, `163885b`, `aa13ce3` and `fb7e7c7`, the scoring scripts are in `d5ebab4`, and the added forest scores are in `d020079`, later than the rest. All commits are dated 13 November 2026. The lake amendment and the probe design were committed together with their results (`5f80de0`, `a26c3e0`), so their order is recorded in the files, not by commit order. Sentinel data are free and open (Copernicus); Hansen and Hi-MAG are CC-BY 4.0; Prithvi-EO-2.0 is Apache-2.0; the SSL4EO weights are CC-BY 4.0.

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